

# Austin Ulrigg

## Curriculum Vitae

Seattle, WA

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austinulrigg.com 

### Education

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#### University of Washington

September 2023 – June 2026

*B.Sc. in Mathematics, Departmental Honors*

*Seattle, WA*

- GPA: 3.83/4.00

### Publications / Preprints

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#### On The Genus Polynomial of Cubic Graphs

2026

*Independent Research*

*Seattle, WA*

- Proved the genus polynomial of a cubic graph is a cycle matroid invariant and incomparable with the adjacency spectrum, constructed an infinite family of co-spectral pairs whose minimum genera differ.
- Computed the genus polynomials of all 7,875,918 connected cubic graphs through 22 vertices.
- Submitted to *Discrete Mathematics*, 2026. [arXiv:2607.25410](https://arxiv.org/abs/2607.25410).

#### An Efficient Genus Algorithm Based on Graph Rotations

Spring 2024–Winter 2025

*Joint work with Alexander Metzger*

*Seattle, WA*

- Developed a novel algorithm to compute the orientable genus of graphs in  $\mathcal{O}(n(4^m/n)^{n/t})$ .
- Successfully determined the exact genus of several graphs for which it was previously unknown.
- Published in *Discrete Mathematics* 349 (2026), 115308.  
[doi:10.1016/j.disc.2026.115308](https://doi.org/10.1016/j.disc.2026.115308). [arXiv:2411.07347](https://arxiv.org/abs/2411.07347).

### Research Experience

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#### Undergraduate Honors Thesis

Fall 2025

*Supervisor: François Clément*

*University of Washington, Seattle*

- Authored a comprehensive survey on graph embeddings on surfaces, rotation systems, and genus algorithms (including PAGE).
- Investigated the classification of forbidden toroidal minors and related open problems.
- Full text available: [austinulrigg.com/post-thesis.html](https://austinulrigg.com/post-thesis.html)

#### Undergraduate Research in Topological Graph Theory

Summer 2025–Spring 2026

*Advisor: François Clément*

*Seattle, WA*

- Worked on a project on classifying forbidden minors for the torus, specifically for graphs excluding a  $K_5$  minor.

#### Washington Experimental Math Lab Researcher

September 2024–December 2024

*Advisor: Professor Stefan Steinerberger*

*Seattle, WA*

- Investigated the asymptotic behavior of the dynamical system described in [arXiv:2409.08961](https://arxiv.org/abs/2409.08961)

#### Washington Experimental Math Lab Researcher

March 2024–June 2024

*Advisor: Professor Hadrian Quan*

*Seattle, WA*

- Collaborated with a team under Dr. Hadrian Quan to investigate wave propagation on graphs, focusing on spectral analysis of the graph Laplacian, graph products, and path homotopy.

## *Presentations*

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>A Practical Genus Algorithm</b>                                                                                                                                          | Nov 2025             |
| <i>Northwest Undergraduate Mathematics Symposium (NUMS)</i>                                                                                                                 | <i>Bothell, WA</i>   |
| • Contributed talk based on joint work with Alexander Metzger.                                                                                                              |                      |
| <b>A Practical Genus Algorithm</b>                                                                                                                                          | May 2025             |
| <i>UW Undergraduate Research Symposium</i>                                                                                                                                  | <i>Seattle, WA</i>   |
| • Presented a new approach to genus computation using an $\mathcal{O}(n(4^m/n)^{n/t})$ algorithm, emphasizing theoretical advancements and impacts on computational fields. |                      |
| <b>Euler Characteristic of a Torus and the Utilities Problem</b>                                                                                                            | July 2024            |
| <i>Centralia High School</i>                                                                                                                                                | <i>Centralia, WA</i> |
| • Talk for $\sim 30$ advanced high-school students on surfaces of various genera, Euler's formula, and gave a solution of the utilities problem on a torus.                 |                      |

## *Honors and Awards*

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|-----------------------------------------------------------------------|--------------|
| <b>Husky 100</b> (Nominated by Dept. of Mathematics)                  | 2025         |
| <b>Math Alliance Predoctoral Scholar Nominee</b>                      | 2024         |
| <b>Phi Theta Kappa Honor Society</b> (International Honor Society)    | 2021–Present |
| <b>Outstanding Student of the Year Nominee</b> (Centralia College)    | 2022         |
| <b>High School Valedictorian</b> (Rank 1/200+, Centralia High School) | 2022         |
| <b>Merit Scholarships Awarded</b> (Total: \$12,000)                   | 2022         |

## *Professional Experience*

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| <b>Mathematics Domain Expert (AI Training)</b>                                                                                                                                                                        | Aug 2023 – July 2024; Aug 2025 – Present |
| <i>Outlier AI / DataAnnotation</i>                                                                                                                                                                                    | <i>Remote</i>                            |
| • Author and edit complex mathematical prompts and solutions to refine Large Language Model (LLM) reasoning capabilities. Develop and apply rubrics to assess model performance and categorize content by difficulty. |                                          |
| <i>Outreach and Service</i>                                                                                                                                                                                           |                                          |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Reading Group Organizer</b>   <i>University of Washington</i>                                                                                                                    | 2023 – 2025          |
| • Co-organized and led reading groups in topology, algebra, linear algebra, and complex analysis for 70+ total students, facilitating discussions, grading, and assigning problems. |                      |
| <b>Teaching Assistant (MATH 209)</b>   <i>University of Washington</i>                                                                                                              | Sep 2024 – Mar 2025  |
| • Assist with ODEs/PDEs (heat, wave, etc.); grade assignments, provide one-on-one support.                                                                                          |                      |
| <b>Mathematics Tutor (CLUE)</b>   <i>University of Washington</i>                                                                                                                   | Sep 2024 – June 2025 |
| • Tutor courses from introductory calculus to abstract algebra.                                                                                                                     |                      |
| <b>Teaching Assistant</b>   <i>Husky Math Academy</i>                                                                                                                               | Oct 2025 – June 2026 |
| • Provide instruction, building problem-solving skills beyond the standard curriculum.                                                                                              |                      |
| <b>Math Student Council</b>   <i>University of Washington</i>                                                                                                                       | 2024 – 2026          |
| • Council Member: Organize study halls and events; act as liaison between students and faculty.                                                                                     |                      |
| <b>VP &amp; Recruitment Chair</b>   <i>Theta Delta Chi Fraternity</i>                                                                                                               | 2024 – 2026          |
| • Elected officer for a 70+ member chapter; oversee recruitment and chapter operations.                                                                                             |                      |
| <b>Online Mathematics Community</b>   <i>Volunteer Tutor</i>                                                                                                                        | Jan 2023 – Present   |
| • Authored 100,000+ posts in a community of 200k+ members, facilitating math discussions.                                                                                           |                      |
| <b>Mathematics Tutor</b>   <i>Mathnasium</i>                                                                                                                                        | July 2025 – Sep 2025 |
| • Tutored K–12 students, focusing on foundational skills and confidence building.                                                                                                   |                      |
| <b>Solution Notes: Pearls in Graph Theory</b>   <i>Independent Project</i>                                                                                                          | May 2024 – June 2024 |
| • Authored a 12-page solution set for Hartsfield–Ringel's <i>Pearls in Graph Theory</i> . [View Notes]                                                                              |                      |
| <b>STEM Tutor</b>   <i>Centralia College</i>                                                                                                                                        | May 2020 – June 2022 |
| • Tutored Calculus I–III, Statistics, and Human Biology; supported center operations.                                                                                               |                      |
| <b>Summer Credit Retrieval Assistant</b>   <i>Centralia High School</i>                                                                                                             | Summers 2017 & 2018  |
| • Assisted high school students with recovering mathematics and science credits.                                                                                                    |                      |

## *Graduate Coursework*

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Graph Theory (Math 563), Modern Algebra I–III (Math 504–6), Linear Algebra (Math 582–3).

## *Other Interests*

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**Hobbies:** Intramural Volleyball (especially beach), Hiking, Dogs, Powerlifting, Reading